University of the Philippines – Open University
Master of Information Systems

System Requirements Specification (SRS) Document

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ExamInsight: ATTAINMENT AND PROGRESS TRACKING IN YEAR 2 INTERNAL ASSESSMENTS AND EXTERNAL BENCHMARK TEST AT PRISTINE PRIVATE SCHOOL

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IS 295a | Information Systems Project I

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Review & Approval

Requirements Document Approval History

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1. Introduction

1.1 Introduction

The purpose of this document is to define and describe the requirements of the project or proposed system, and to spell out the functionalities and constraints including its system features.

1.2 Scope of this Document

This document encompasses the project's objectives, scope, both functional and non-functional requirements, as well as details regarding the required software and hardware. Additionally, it provides insights into the project's environmental considerations, including security requirements and criteria for software quality.

The designated users for this proposed project are the Head of School, Deputy Head of School and Year 2 class teachers at Pristine Private School – Junior Campus, located in Dubai, United Arab Emirates. The developer of the system is Aldwin C. Sumalinog, who will be working independently in accordance with the University of the Philippines – Open University's Master of Information Systems program requirements. One of the identified constraints includes the deadline for the paper which is due on 18 May 2024.

1.3 Overview

The project is called *ExamInsight: Attainment and Progress Tracking in Year 2 Internal Assessment and External Benchmark Tests at Pristine Private School*. It will encompass four essential operations crucial for establishing and overseeing data, which are indispensable for implementing persistent storage applications: creating, retrieving, updating, and deleting. The system will utilize the Python programming language for coding, building models and predicting results. To generate dashboards and present data in the most simple and effective way, the system will make use of Dash, an open-source app-building framework for Python. This will be complemented with Matplotlib paired with Pandas and Numpy for basic graph plotting and exploratory data analysis, and Seaborn for statistics visualizations.

1.4 Business Context

Pristine Private School, an award-winning UK Curriculum school established in 1992, caters to a highly diverse student body of over 47 nationalities from Foundation Stage 1 to A-Level (Year 13). Academic year 2022-2023, Pristine Junior Campus opened a new state-of-the-art school facility purpose-built for the learning needs of the youngest minds (FS1 to Y2). This new campus offers modern-day learning spaces through a bespoke design for a vibrant learning environment that inspires curiosity and creativity, as well as stimulates intellectual growth in children [1].

2. General Description

2.1 Product Functions

The system is designed to automate the tracking of year 2 students' attainment and progress in internal assessments and external benchmark tests at Pristine Private School. This eliminates the need for the conventional method of using formulas and spreadsheets together, which often lacks in-depth analysis and interpretation of assessment data. Additionally, the system will integrate AI (artificial intelligence) technology, leveraging large language models like ChatGPT, for formative feedback based on assessment data and student performance.

2.2 Similar System Information

Within the current market landscape, numerous analytics platforms/systems are available, often containing advanced features that come with a higher price tag. However, these systems may fall short in providing the customized features essential to meet the specific requirements of an analytics platform/system for tracking attainment and progress in year 2 internal assessments and external benchmark tests at Pristine Private School – Junior Campus. As a result, the school has chosen to pursue a solution that better aligns with its distinct requirements. The possible strength of the proposed system is its innovation value of integrating AI technology to generate detailed student profile description and provide in-depth data analysis and interpretation for cohort and year group assessment data.

2.3 User Characteristics

The system primarily caters to the head of school, deputy head of school, and year 2 homeroom or class teachers with varying levels of computer proficiency. It is intuitively designed to facilitate straightforward navigation and access to content, requiring minimal onboarding for users ranging from basic to intermediate computer literacy

2.4 User Problem Statement

Teachers and administrators, including head of school and deputy head of school, have observed that tracking student attainment and progress in year 2 internal assessments and external benchmark tests is time-consuming and lack comprehensive analysis capabilities when done in the current conventional method utilizing manual data entry and spreadsheet systems. There is a need for a user-friendly system that centralizes data, streamlines and automates tracking of attainment and progress in a given assessment data, and offers insightful analytics on students' performance using AI technology like ChatGPT. The proposed system features analytics dashboards with formative feedback, enabling the identification of areas to improve, gaps in knowledge, support resources needed, and learning strategies to adapt.

2.5 User Objectives

The school administrators (head of school, deputy head of school) of Key Stage 1 in Pristine Private School – Junior Campus are keen to implement an automated system that tracks the attainment and progress of Year 2 assessments, both internal and external benchmark tests, including the analysis and interpretation of individual student performance through formative feedback using AI technology (i.e., ChatGPT). The management strongly believes that leveraging such cutting-edge tools for educational advancement will streamline data management, provide comprehensive insights into student performance, and facilitate informed decision-making to support academic growth.

2.6 General Constraints

In compliance with data protection regulations, the system must ensure the security and confidentiality of student assessment data at all times. Additionally, it must be compatible with existing hardware and software infrastructure used by Pristine Private School – Junior Campus. Finally, the system shall be scalable to accommodate potential increases in the volume of assessment data, aligning with anticipated increases in enrollment over time.

3. Functional Requirements

3.1 General Functional Requirements

These are the general functional requirements of the proposed system.

3.1.1. **User Authentication and Role-Based Access Control**. The system shall provide secure authentication mechanisms, including username/password login and role-based access control, to ensure authorized access to assessment data based on user roles (e.g., administrators, teachers).

3.1.2. **Data Management and Privacy**. The system shall prioritize the efficient creation, secure storage, retrieval, and management of assessment data, including student profiles, test scores, and progress category indicators, while complying with relevant data privacy regulations to safeguard student privacy and ensure legal compliance in handling assessment data and other student-related information.

3.1.3. **Analytics and Reporting**. The system shall provide comprehensive analytics through dashboards and reporting functionalities for teachers and school administrators, allowing them to gain insights into student performance trends, identify areas for improvement, and track overall academic progress.

3.1.4. **AI Technology Integration**. The system shall implement AI-powered data analysis capabilities, leveraging large language models like ChatGPT, to enable automatic interpretation of assessment data and provide valuable insights to support informed decision-making and academic planning.

3.2 Specific Functional Requirements

The following are the proposed system's specific functional requirements, along with its description of each feature:

3.2.1. **Login Facility**. The system shall allow secure login with username and password authentication for teachers and school administrators to ensure access control and protect assessment data.

3.2.2. **Logout Facility**. Users shall be able to logout from the platform by clicking the logout button.

3.2.3. **Accounts and Access Control Facility**. The system shall allow admin users to manage user accounts and grant or restrict access to system features and data, tailored to user privileges.

3.2.4. **Import External Data Facility**. The system shall allow users to seamlessly import assessment data from external sources, and provide compatibility with standard file formats (e.g., CSV, Excel) for data integration.

3.2.5. **Reports and Analytics Facilities**. The system shall enable users to generate comprehensive reports on student performance metrics and academic trends for attainment & progress. Additionally, it shall offer advanced analytics to visualize data and derive actionable insights.

3.2.6. **Dynamic Dashboards**. Users shall have access to a centralized interface through dashboards to gain valuable insights from both internal and external benchmark assessment data, as well as student performance metrics to gauge academic progress and attainment.

3.2.7. **Intelligent Data Analysis Facility.** The system shall leverage AI-technology to automatically interpret assessment data, trends and patterns in student attainment & progress, offer formative feedback to address knowledge gaps, areas to improve, support resources needed and learning strategies to adapt, as well as offer actionable insights for teachers.

4. Interface Requirements

4.1 User Interfaces

The system shall feature an intuitive user interface with easy navigation and user-friendly controls, ensuring accessibility and usability for users with varying levels of technical expertise. User interface elements, from buttons to inputs, navbars, alerts, or cards, will be developed using a web UI toolkit like Bootstrap or Materialize CSS to ensure responsive design and consistent UI components. Analytics dashboards will be constructed using Dash, an open-source app-building framework for Python with libraries such as Pandas, NumPy, and SciPy for data manipulation and statistical analysis. Furthermore, HTML, CSS, and JavaScript will be used for UI development, with JavaScript libraries and Vue.js framework utilized to create interactive front-end components.

4.2 Hardware Interfaces

In lieu of hardware components, the system shall be hosted on a cloud-computing platform like AWS EC2 (Elastic Cloud Compute) to run the web application. For development and testing purposes, the system shall be deployed on Amazon EC2 Free Tier with 750 hours per month for Linux, RHEL and Windows t2.micro/t3.micro, and public IPv4 address. [2]

4.3 Communications Interfaces

The system shall utilize Flask-RESTful API to facilitate communication between the system's backend and front-end applications. It will employ encryption for data before transmission and decryption upon receipt to ensure secure transmission during API calls. Tools like Postman or Insomnia may be used for testing API end points and network requests. Furthermore, the system shall integrate the OpenAI API for ChatGPT to enhance the system's AI capabilities, elevating its innovation and functionality. Additionally, networking protocols like HTTPS shall be used for secure communication over the Internet.

4.4 Software Interfaces

The system shall use an open-source Python framework called Flask for building the backend logic and API end points. Additionally, Jinja2 will be used to provide dynamic components to the web application like rendering HTML content based on data and any other documents with dynamic content. ReportLab shall be used to provide support for users to download reports in various formats.

Containerization tools like Docker shall be utilized for packaging the application and its dependencies, as well as continuous integration and deployment (CI/CD) tools like GitHub Actions for automating the deployment process.

5. Performance Requirements

5.1 Web Browsers

The proposed system shall be compatible with modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. It shall support the latest versions of these browsers to ensure optimal performance, functionality, and user experience. Additionally, the system should be designed with responsive web design principles to accommodate different screen sizes and resolutions, including desktops, laptops, tablets, and smartphones.

5.2 Cloud Hosting

The proposed system shall be deployed and hosted via cloud computing platform for budget consideration without investing in new infrastructure or licensing new software and on-demand availability of computer system resources. Specifically, Amazon Web Services' Elastic Compute Cloud (AWS EC2) will be utilized during the development phase as it offers free usage tier for 12 months with the following services: [3]

- 750 hours of Amazon EC2 Linux or RHEL or SLES t2.micro instance
- 1 GiB memory
- 32-bit and 64-bit platform support
- 5 GB of Amazon S3 standard storage
- 750 hours of elastic load balancer plus 15GB data processing
- AWS Certificate Manager (SSL/TLS)
- AWS CloudWatch (monitoring cloud resources & applications)

6. Other Non-Functional Attributes

6.1 Security

The proposed system's security measures will be tailored to match the sensitivity of the information it handles. All sensitive data transmitted shall be encrypted using industry-standard encryption algorithms (i.e., Transport Layer Security or TLS) to prevent unauthorized access. It shall also employ HTTPS protocol for all communications between clients and server.

The system shall implement role-based access control (RBAC) to restrict access to system features and data based on user roles and permissions. Authentication mechanisms such as username/password shall be used to verify user identities.

6.2 Scalability

The system shall be scalable to handle increasing volume of assessment data as the school's enrollment grows without compromising system performance. This will be handled through several key mechanisms:

- Amazon S3's distributed architecture and auto-scaling mechanisms shall be setup to automate the continuous splitting of data into partitions based on high request rates and increasing volume of data.
- Elastic Load Balancer shall be setup in AWS EC2 instance to accommodate changes in traffic volume and distribute requests.
- AWS Auto Scaling, *a feature available at no additional charge*, shall be setup through the AWS management console to handle changes in workload demands and automatically adjust capacity based on predefined scaling policies, such as CPU utilization and network traffic.

6.3 Reliability

Reliability in the proposed system shall be ensured through redundant infrastructure components deployed on Amazon S3 for data redundancy and availability, and AWS CloudWatch for continuous monitoring and automated alerting of system health. Additionally, fault-tolerant architecture and load balancing with Amazon EC2 Auto Scaling shall be implemented to maintain system availability and resilience against failures. The system shall also be rigorously tested to guarantee reliability.

6.4 Maintainability

Maintainability in the proposed system shall be employed through several strategies:

- The system shall provide comprehensive documentation, including code comments, API documentation, and system architecture diagrams, to facilitate maintenance of the system.
- The system shall utilize version control through Git to track changes to the codebase and ensure that changes can be rolled back or reverted if necessary.
- The system shall implement continuous integration/continuous deployment (CI/CD) pipelines to automate building, testing, and deployment processes.

6.5 Portability

The system shall be designed to utilize Docker as containerization technology to package the application and its dependencies into lightweight, portable containers, ensuring consistency and compatibility across different environments and platforms.

6.7 Reusability

The system shall be designed to employ design patterns and templates through MVC (Model-View-Controller) architecture and/or CRUD (Create-Read-Updated-Delete) operations to easily reuse and adapt existing code and build new features or modules.

6.8 Application Affinity/Compatibility

The system shall undergo compatibility testing to ensure the application will function correctly across different web browsers and devices. It shall be developed using platform-agnostic languages and frameworks, such as HTML5, CSS3, and JavaScript, to ensure compatibility with various platforms and devices, including desktops, laptops, tablets, and smartphones. Additionally, implementation of responsive design techniques in the development phase shall be employed to optimize the user interface and layout for different screen sizes and resolutions, ensuring accessibility and compatibility across a wide range of devices and screen orientations.

6.9 Resource Utilization

The system shall ensure optimal performance using AWS EC2 Auto Scaling which can dynamically allocate computing resources (CPU & memory) based on workload demands. Moreover, the proposed system shall also employ caching mechanism and data storage optimization using Amazon S3 for static content to minimize resource-intensive operations and improve system responsiveness.

6.10 Serviceability

The system shall utilize Amazon CloudWatch to monitor system metrics, log files, and application performance in real-time. Amazon S3 buckets shall be used to store log files and other data for long-term retention and archival. Additionally, the proposed system shall enable administrators to perform system maintenance, updates, and troubleshooting activities by leveraging remote administration capabilities through AWS Management Console.

7. Operational Scenarios

Scenario A: User Authentication

The web application shall require users (teachers, school administrators, system admin) to authenticate using a secure login process. This will involve entering a username and password, which the system will verify against the stored credentials before granting access.

Scenario B: Data Import and Management

The system shall allow for the importation of existing assessment data including student information from various formats (i.e., CSV, Excel). It will provide tools for data management, including editing, updating, and deleting records.

Scenario C: Reporting and Analytics

The application shall offer comprehensive reporting features for users to generate performance report for individual students or entire classes, selecting specific assessment criteria and time periods for analysis. Reports include graphical representations of student progress and comparative analysis against predefined benchmarks. Generated analytics dashboards with the aid of AI technology (i.e., ChatGPT) shall provide insights into student performance, including the identification of trends and patterns in attainment and progress.

Scenario D: System Maintenance

System updates and routine maintenance shall be scheduled to ensure optimal performance. The system shall enable system administrators to perform backups on data.

Scenario E: User Support and Training

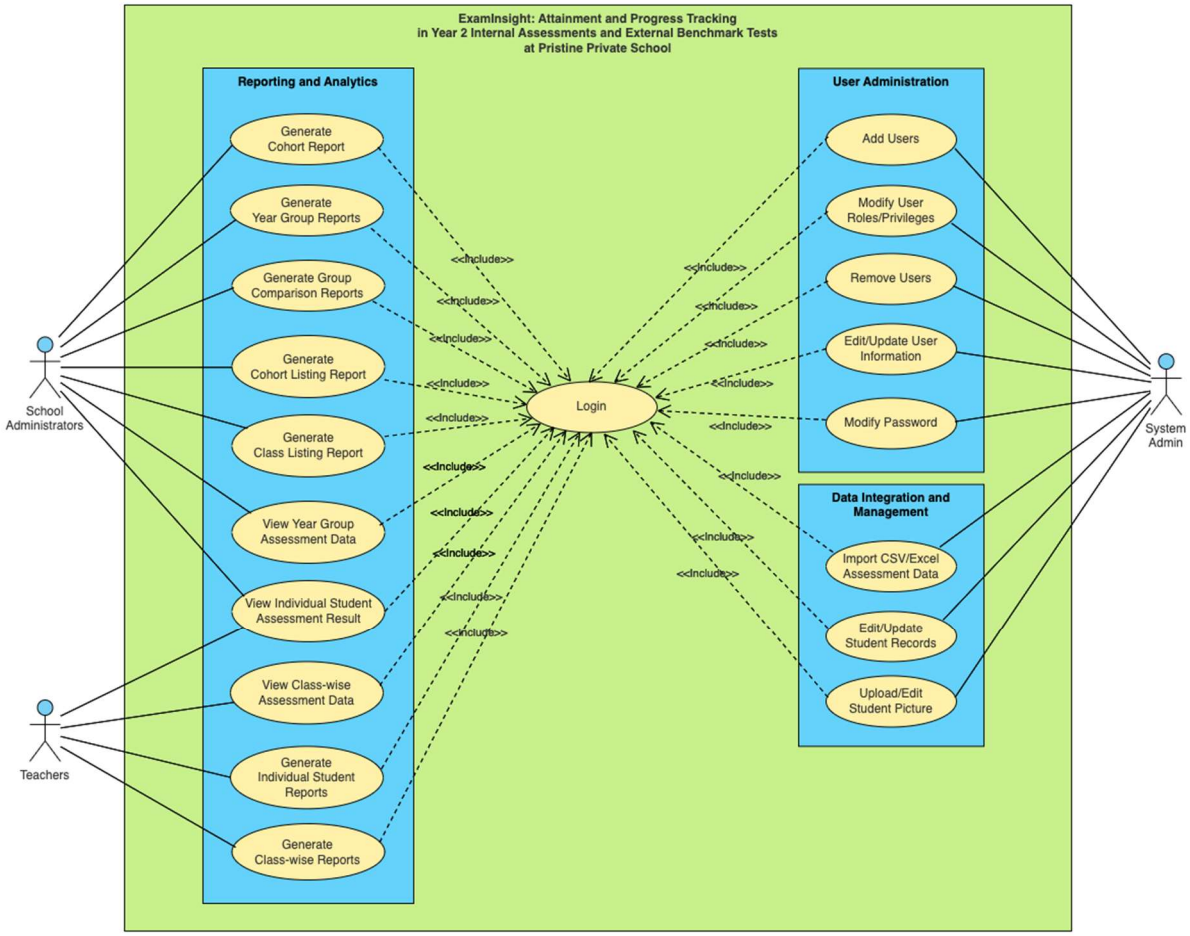
The system shall include training modules to guide users (i.e., teachers and school administrators) on how to use the application for tracking and assessing student progress. Other user support services, including troubleshooting assistance, and training sessions shall be provided to help users navigate the system effectively.

8. Preliminary Use Case Models and Sequence Diagrams

This section outlines the fundamental use cases that fulfill the system's requirements, providing a structured overview of how these requirements can be addressed within the system's framework.

8.1 Use Case Model

Figure 1. Use Case Diagram



The proposed system is designed to accommodate three actors: system administrators, school administrators (head of school/deputy head of school), and teachers. System administrators will have unrestricted access to all system functionalities, including data integration and management, and user administration. On the other hand, school administrators and teachers will have limited access, with permissions tailored to their respective roles and responsibilities. School administrators will have access to most of the system features and services, such as viewing analytics (cohort and/or year group), attainment & progress reports for individual students, and overseeing assessment data. Teachers, as end-users, will have access to class-specific functionalities, including generating progress reports for individual students within their assigned classes, as well as accessing analytics for class-wide attainment and progress. This role-based access control ensures that each user group can perform their designated tasks efficiently while maintaining system security and integrity.

8.2 Sequence Diagrams

Figure 2. Add New Users

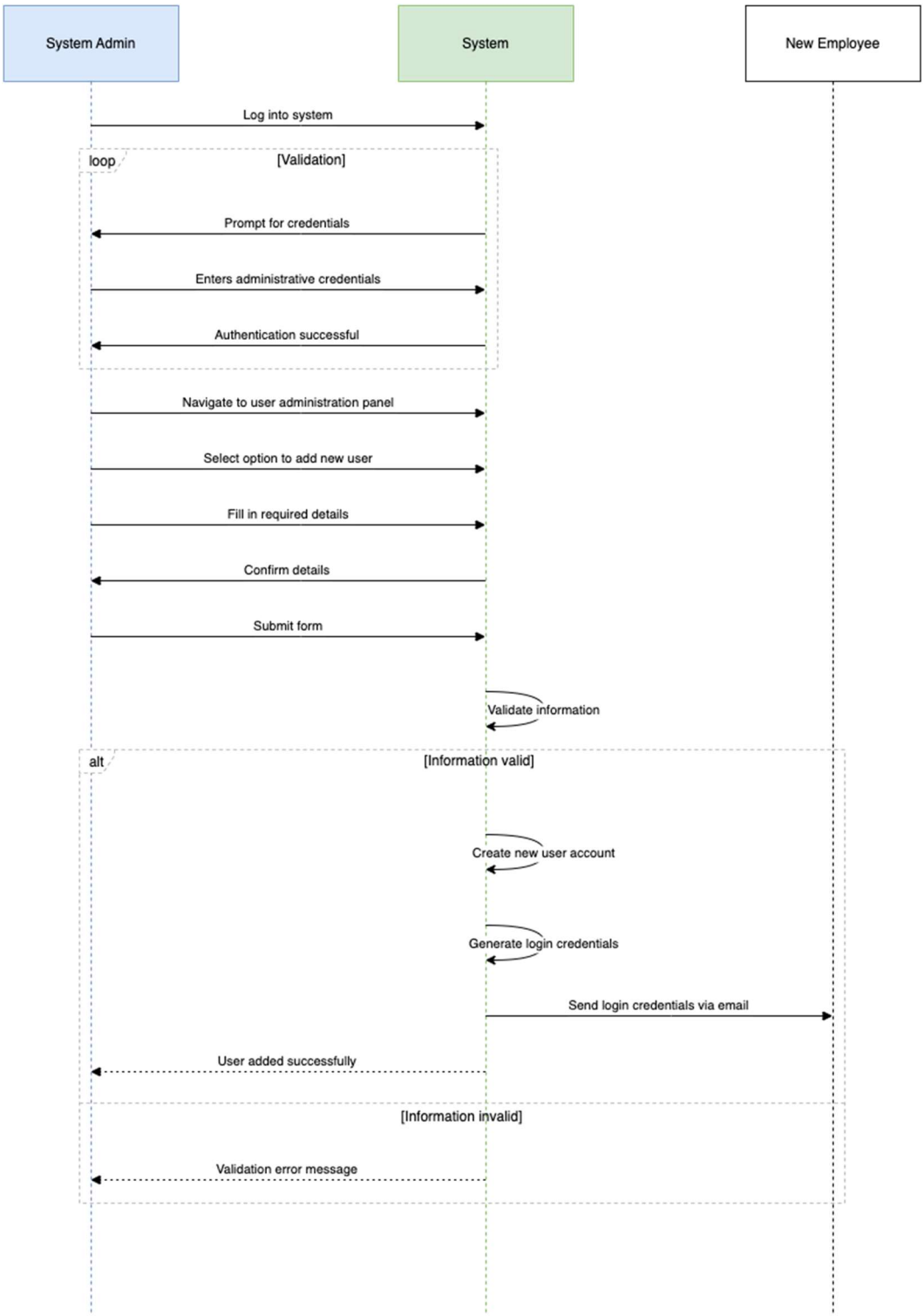


Figure 3. Modify User Role/Privileges

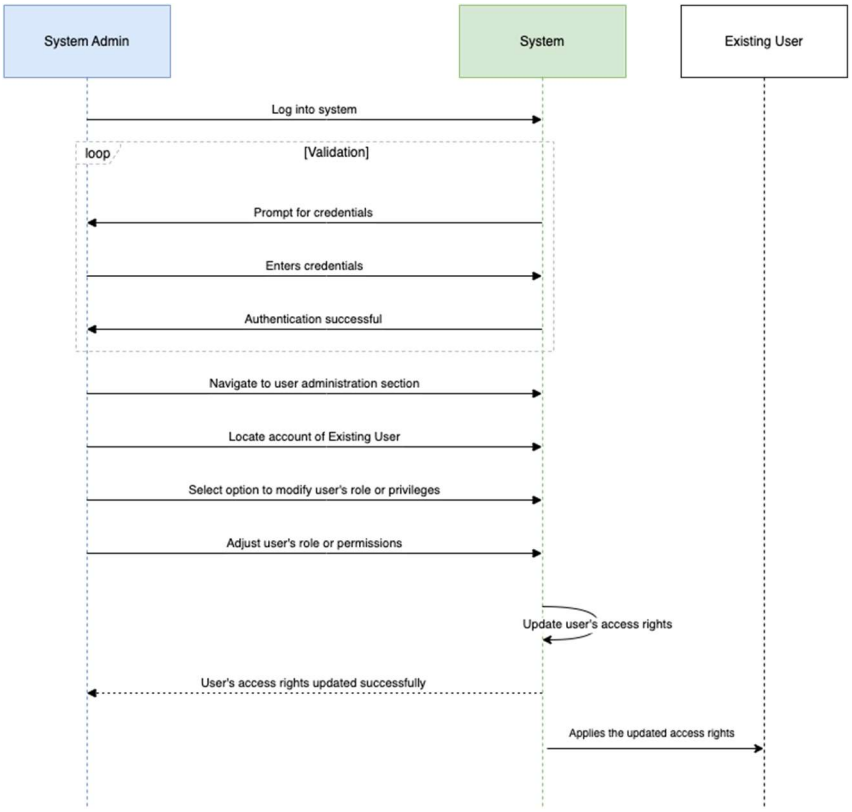


Figure 4. Remove Users

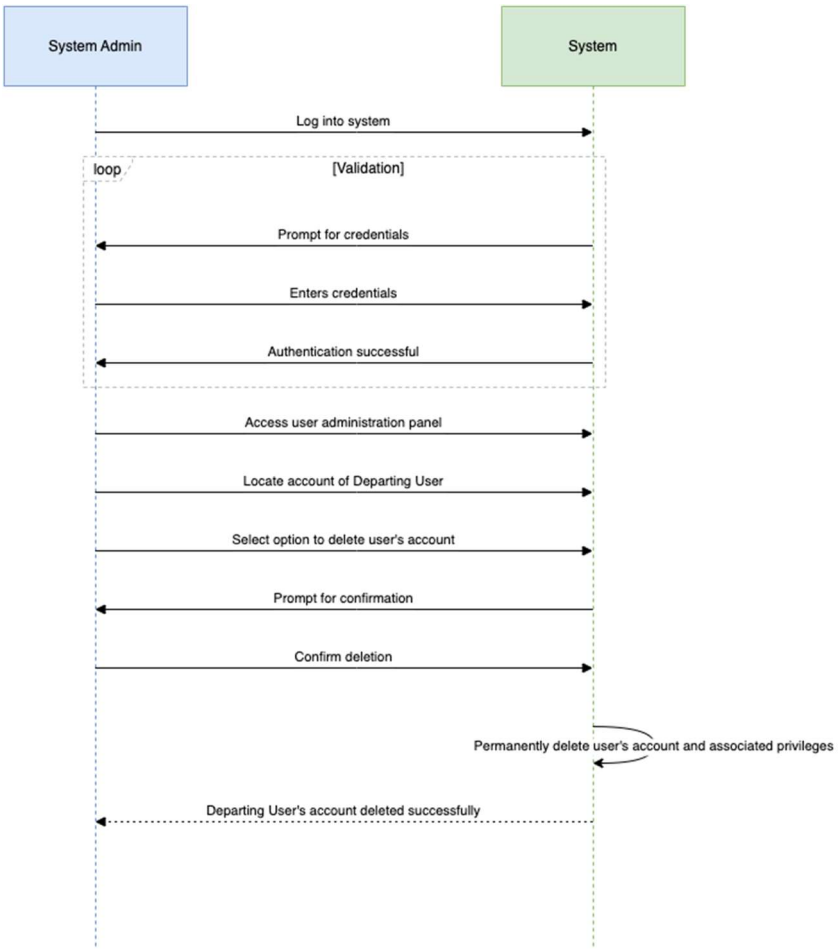


Figure 5. Edit/Update User Information

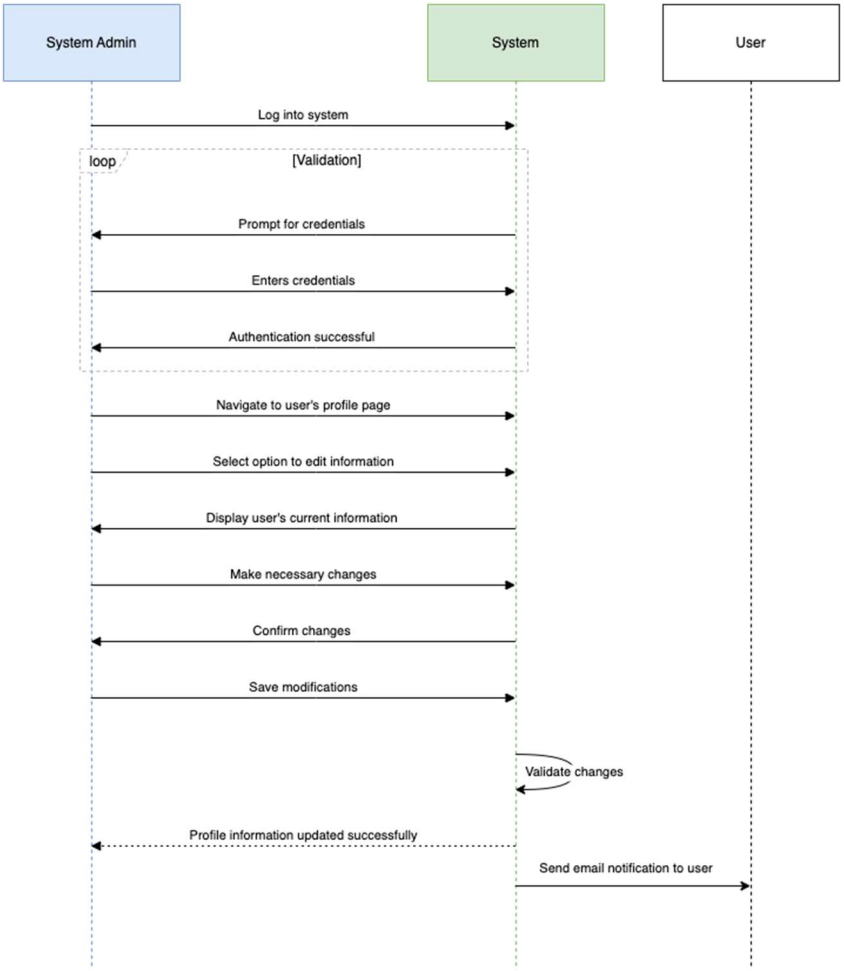


Figure 6: Modify Password

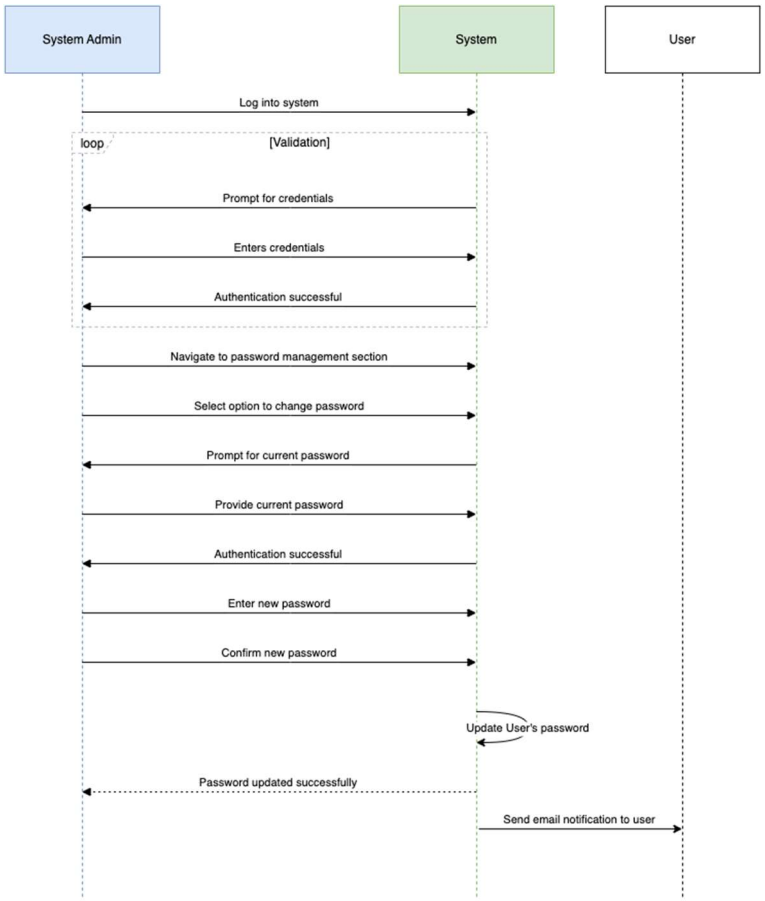


Figure 7. Import Assessment Data

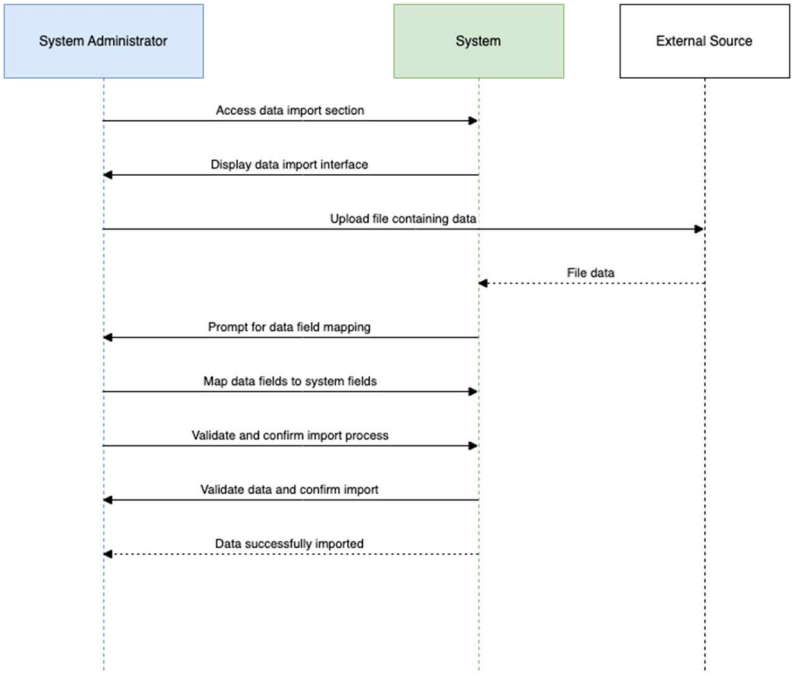


Figure 8. Edit/Update Student Record

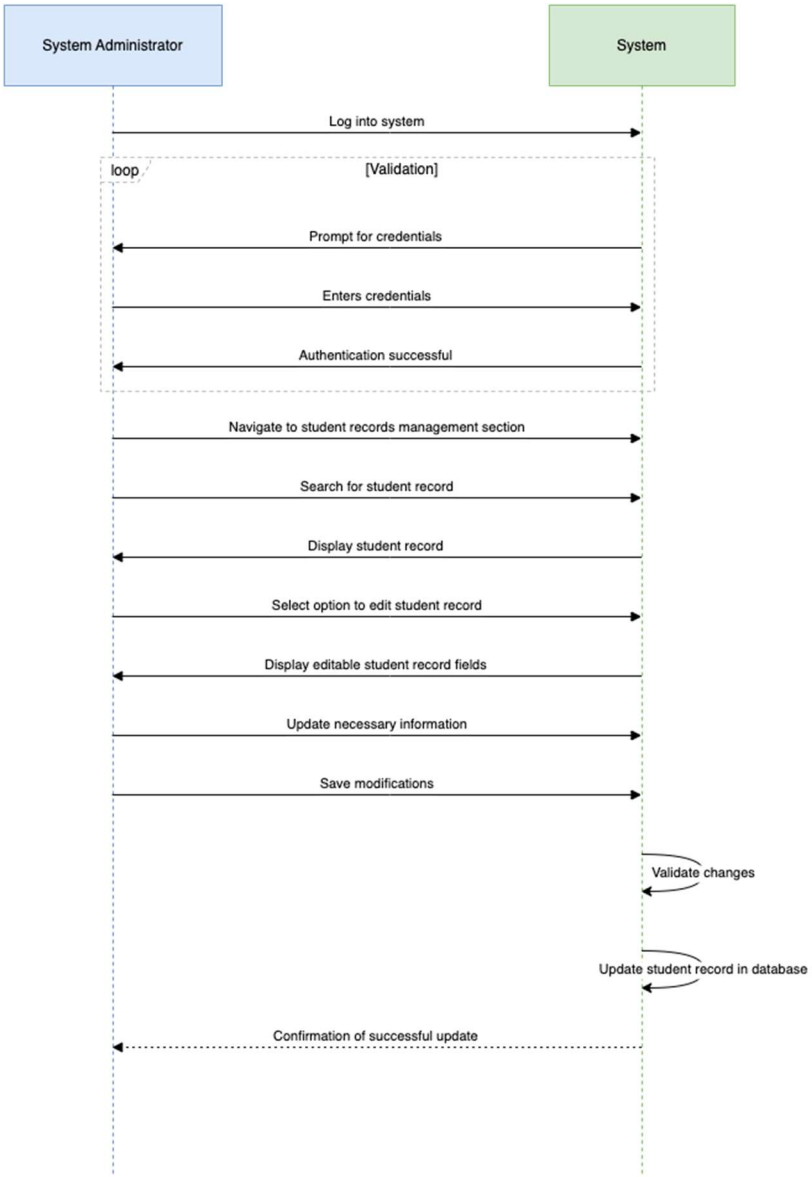


Figure 9. Upload/Edit Student Picture

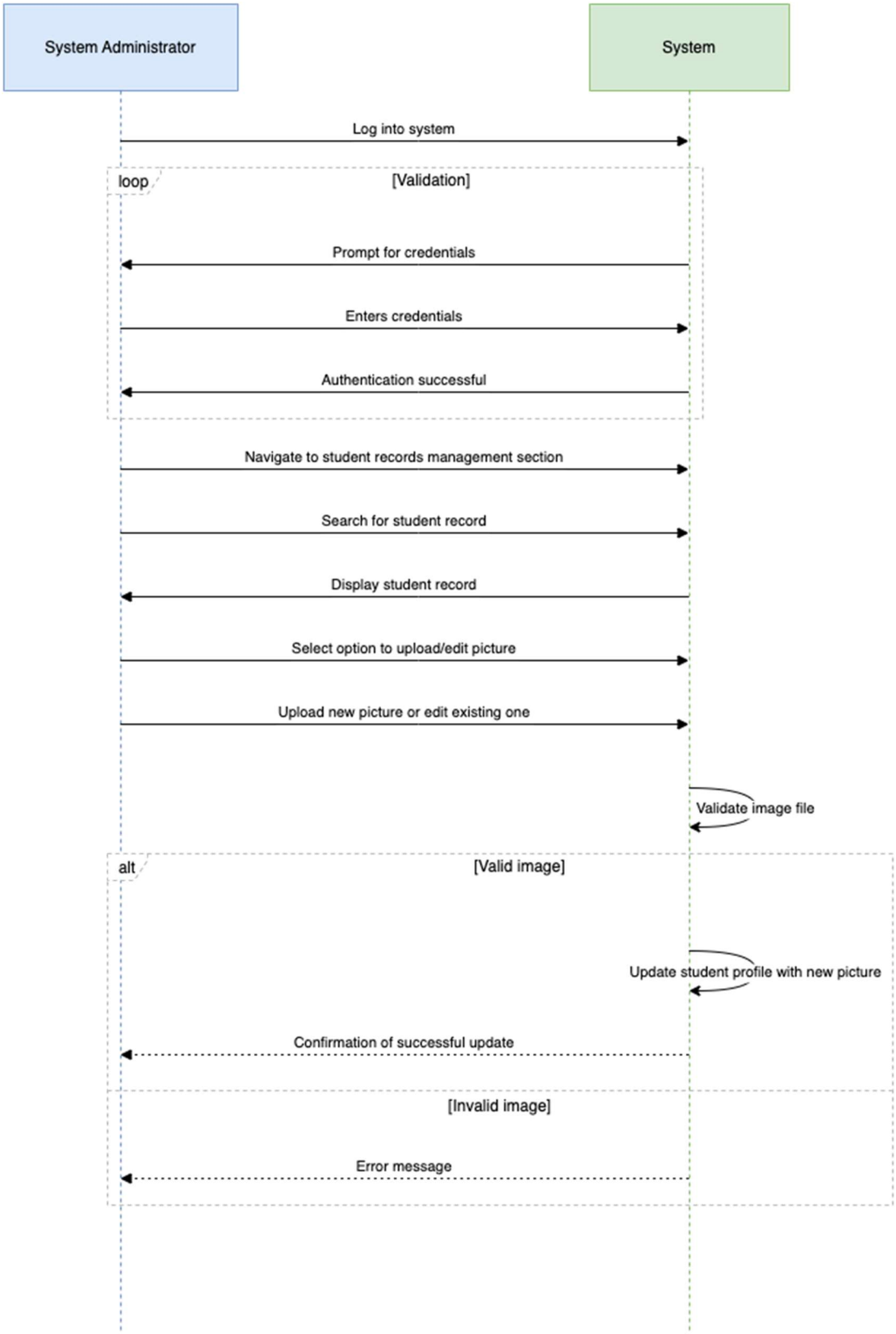


Figure 10. Generate Cohort Report

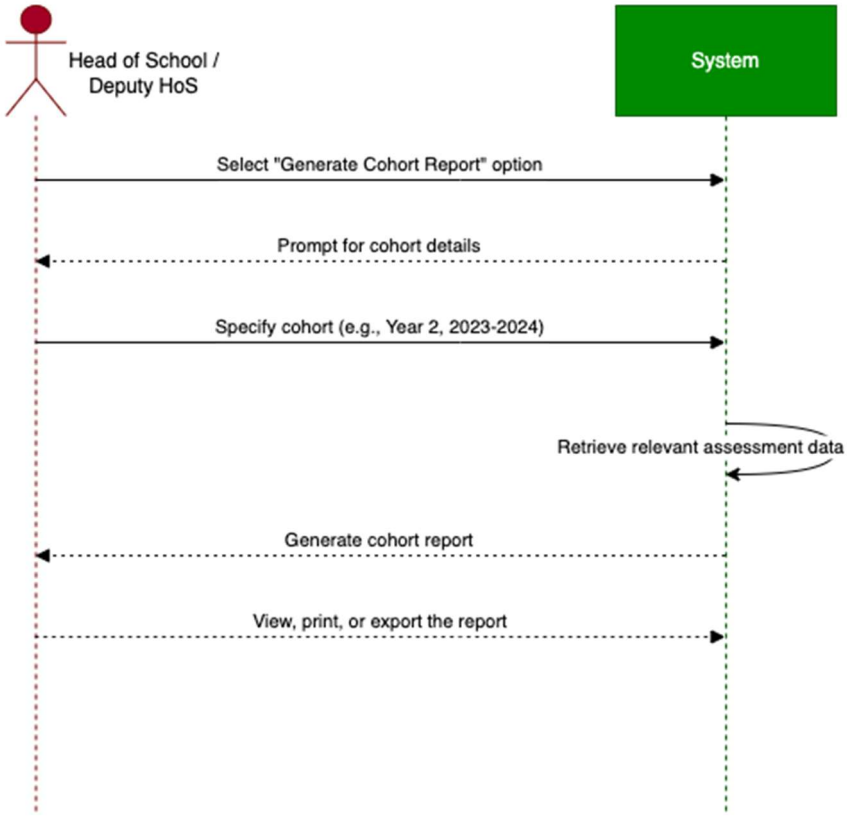


Figure 11. Generate Year Group Reports

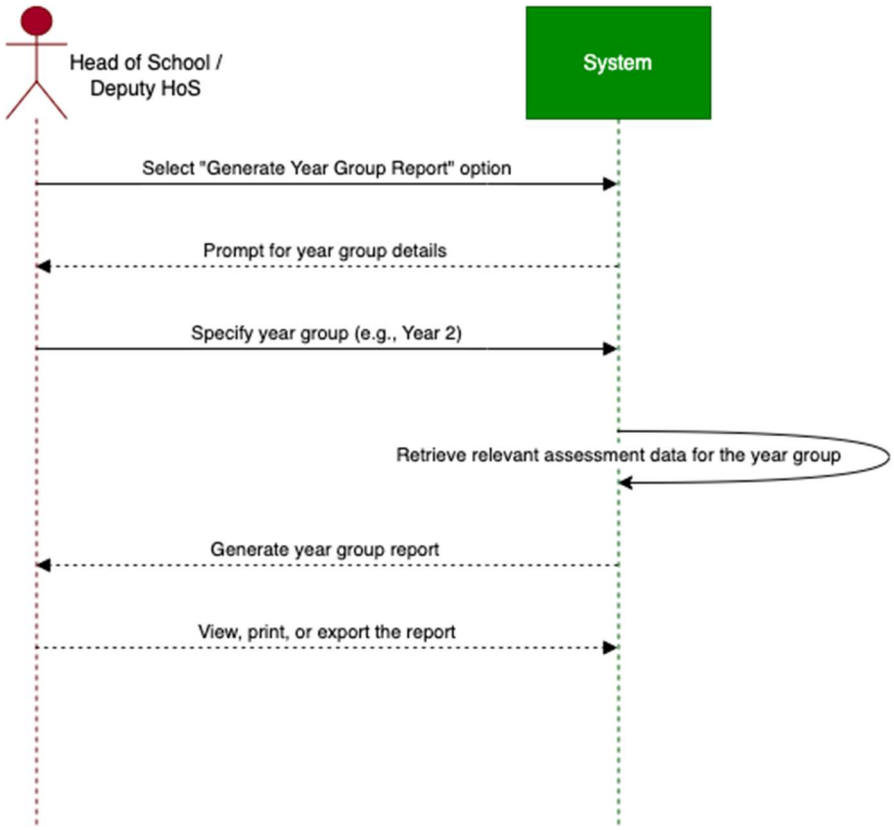


Figure 12. Generate Group Comparison Reports

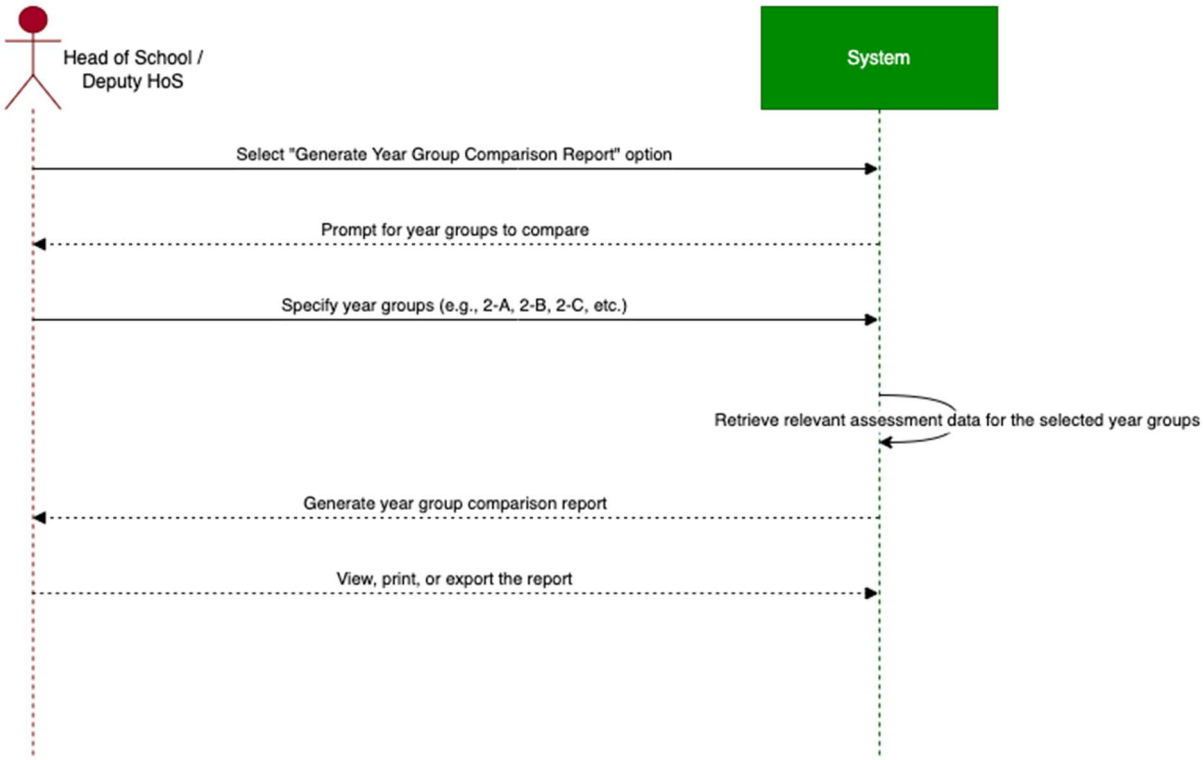


Figure 13. Generate Cohort Listing Report

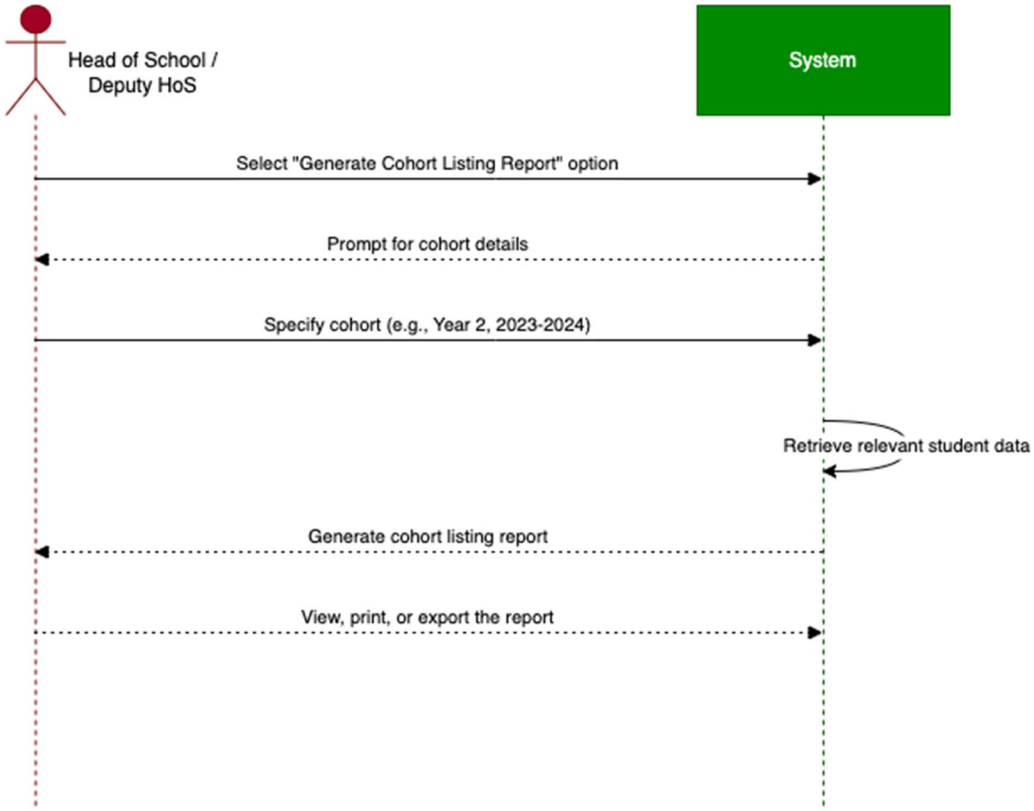


Figure 14. Generate Class Listing Report

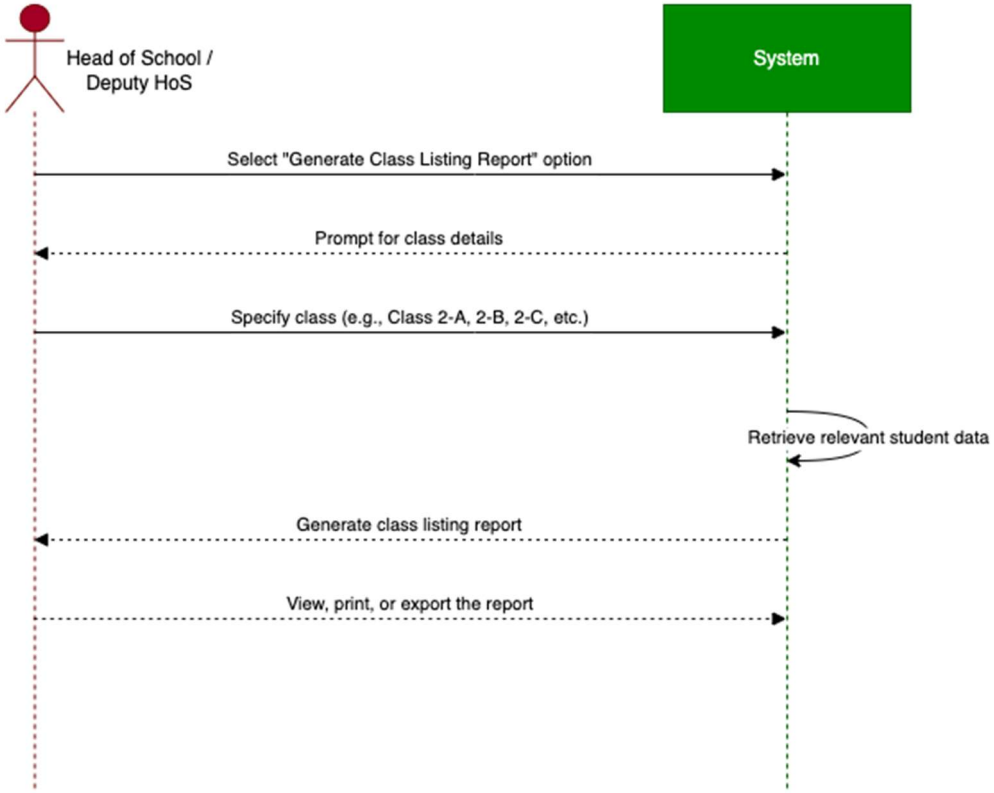


Figure 15. View Year Group Assessment Data

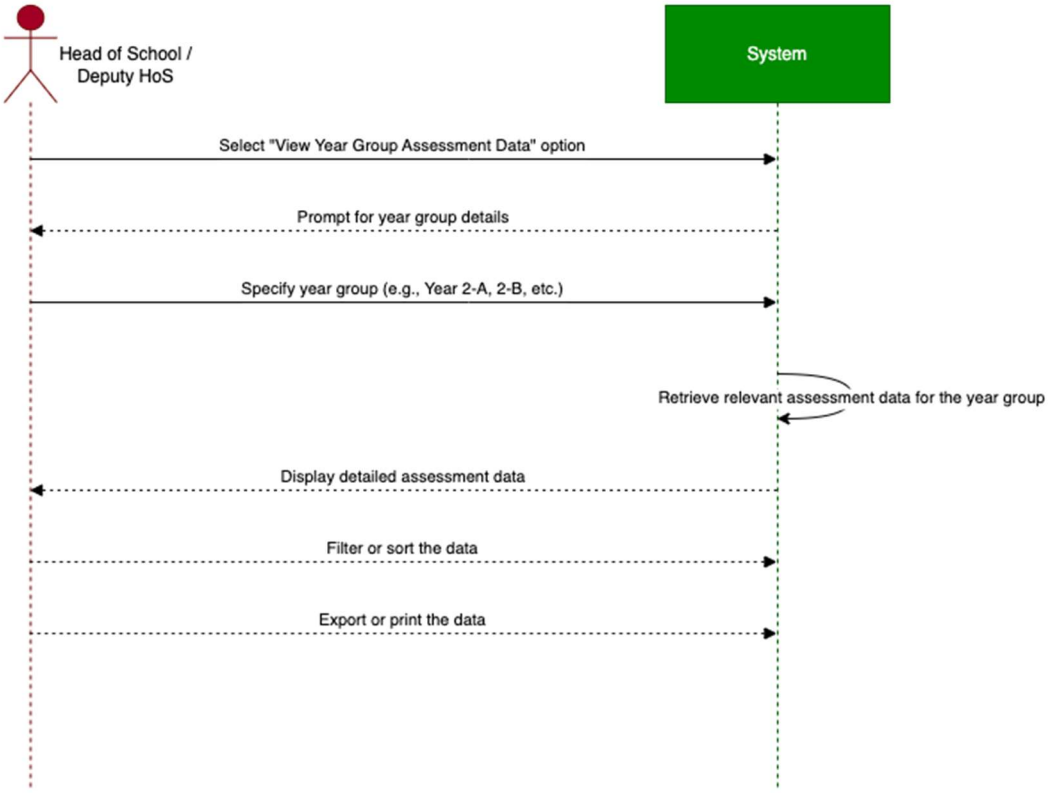


Figure 16. View Individual Student Assessment Result

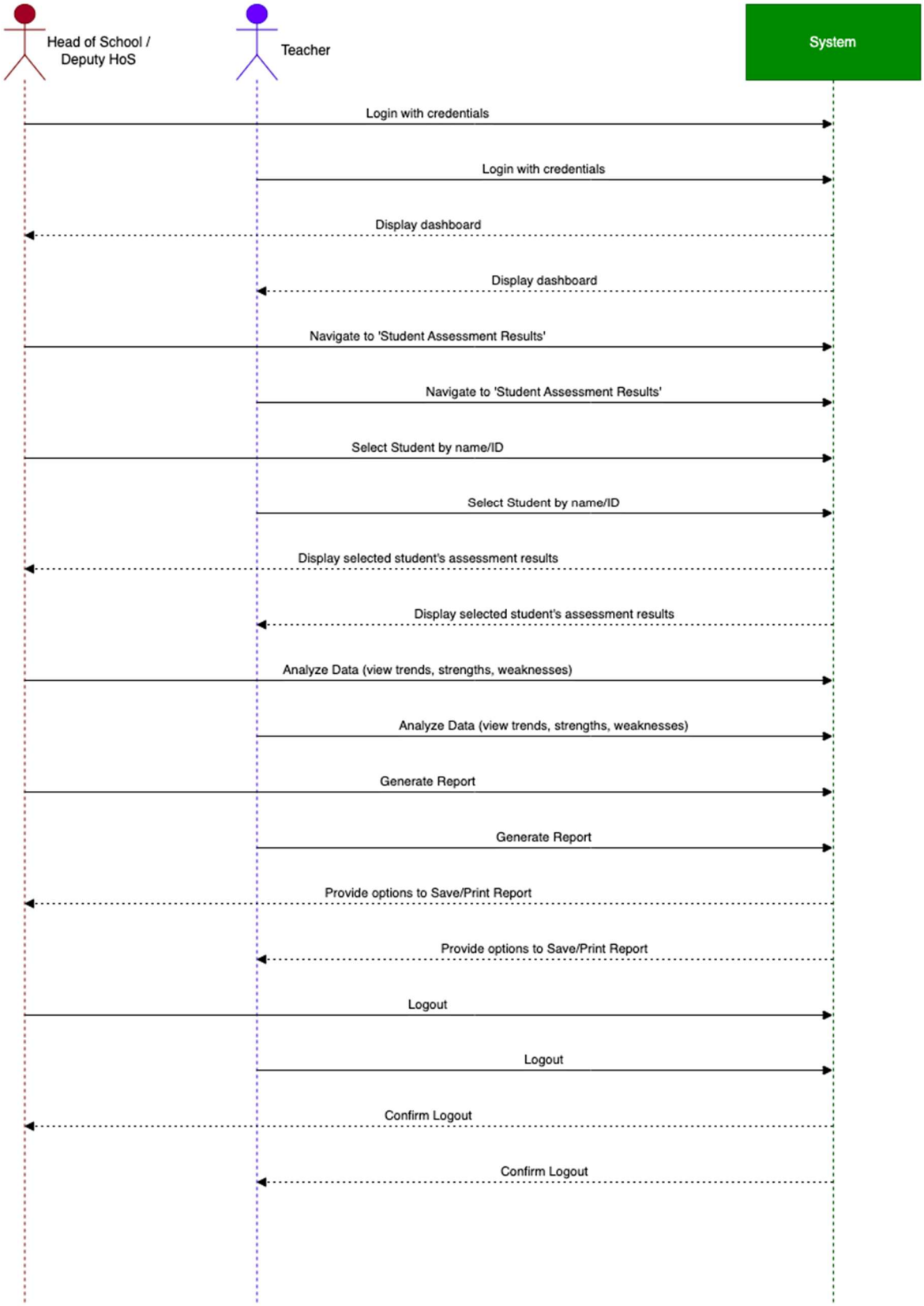


Figure 17. View Class-wise Assessment Data

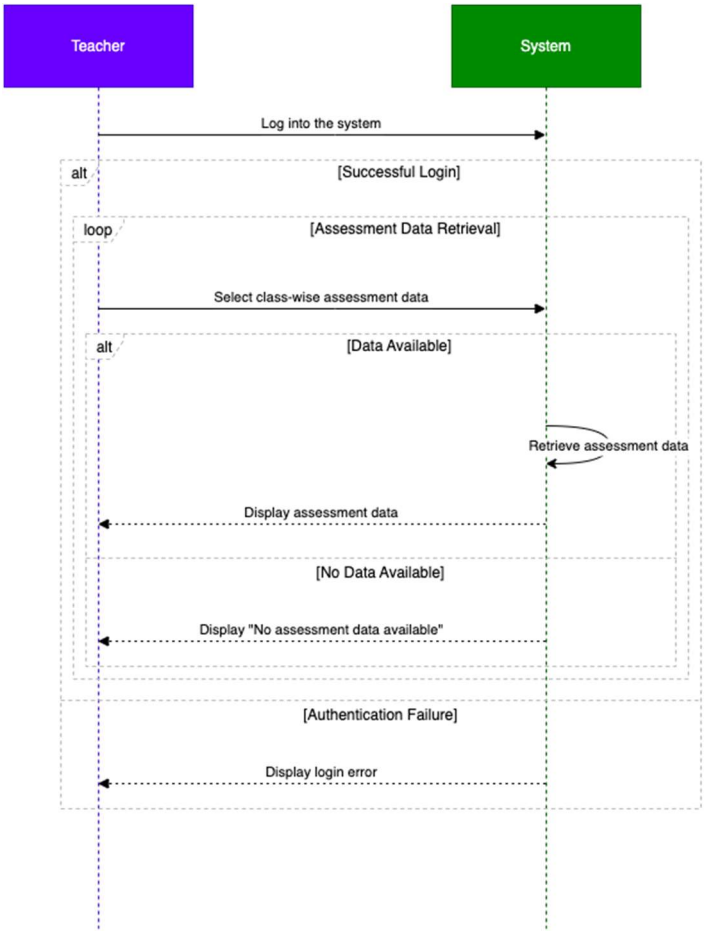


Figure 18. Generate Individual Student Report

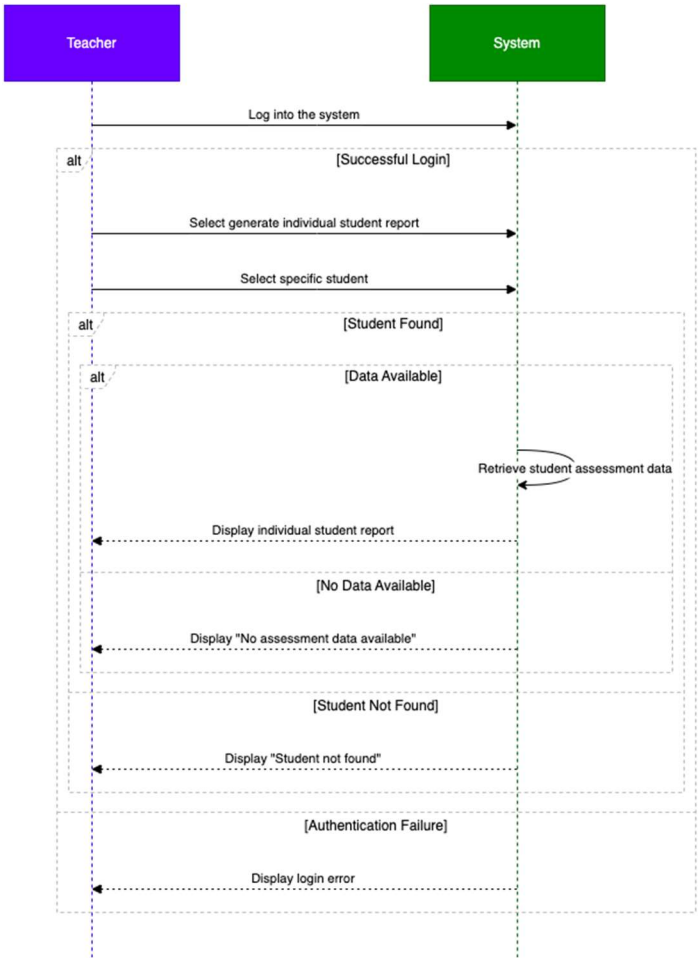
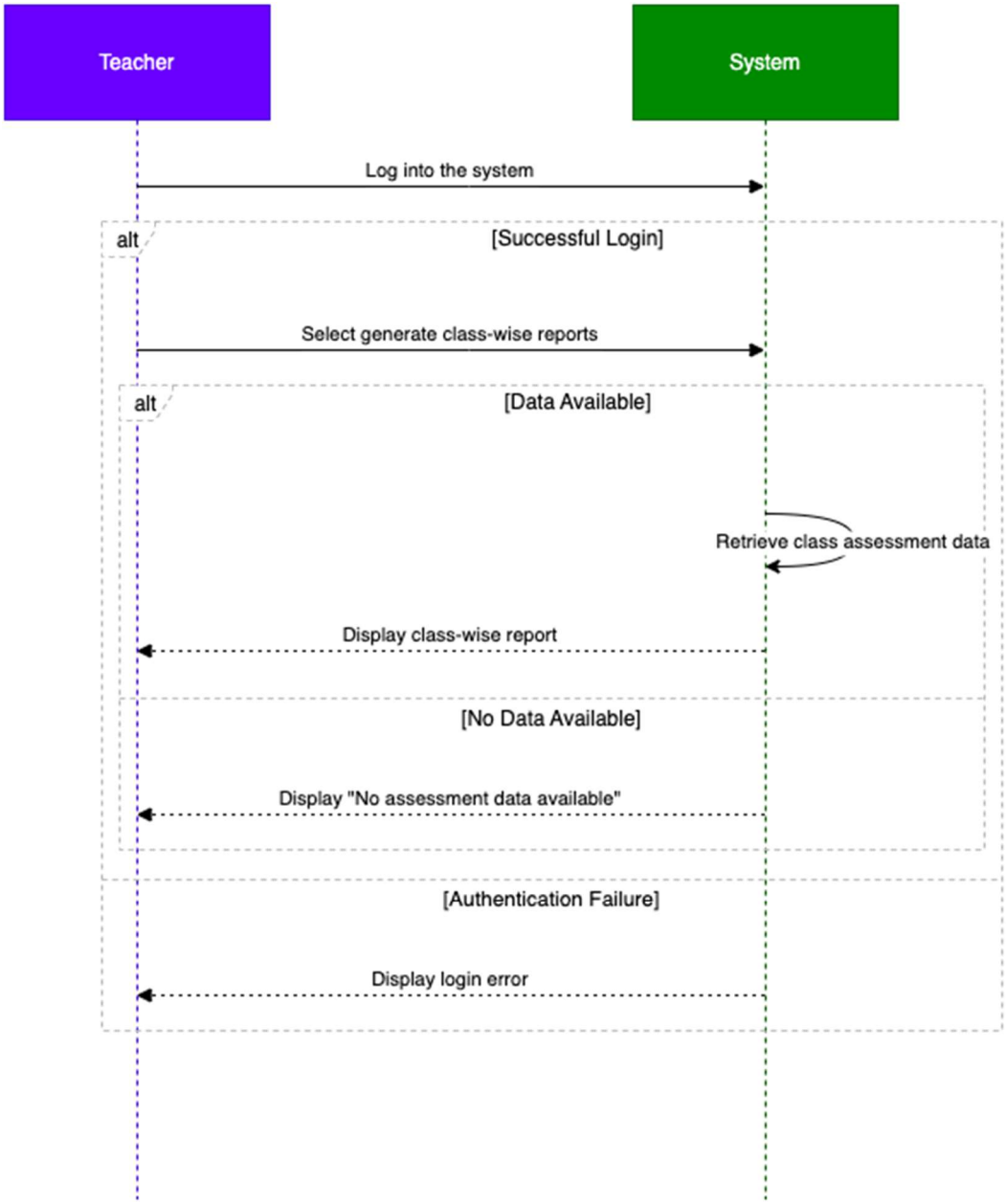
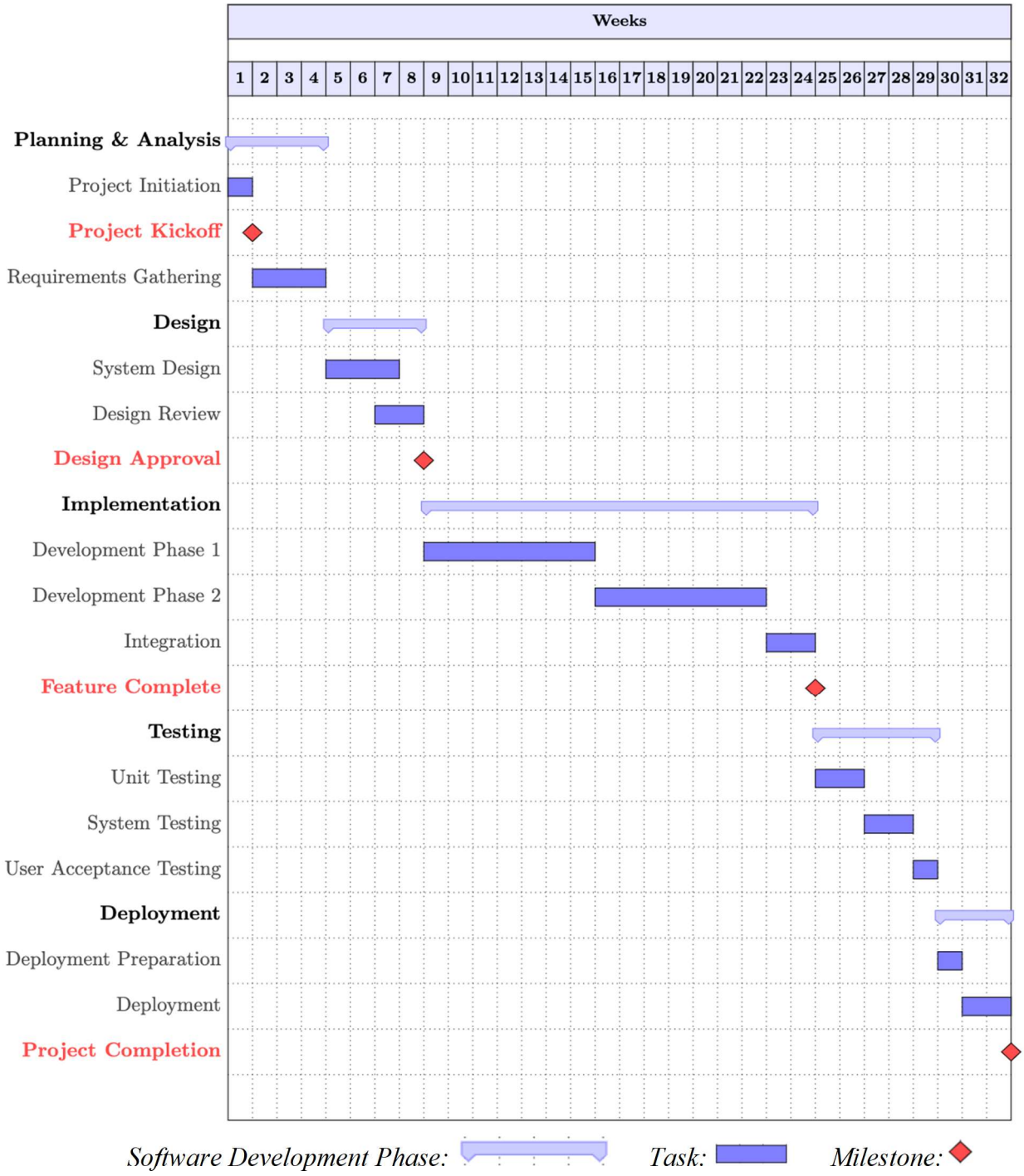


Figure 19. Generate Class-wise Report



9. Updated Schedule



The timeline for this software project involves a comprehensive 32-week plan that ensures a systematic and thorough approach to developing the proposed system, from initial planning to final deployment. Each phase is designed to build upon the previous one, ensuring a smooth transition and progress towards the project's completion.

- **Phase 1: *Planning & Analysis Phase*** (4 weeks): This initial phase is dedicated to planning and analysis, culminating in the *'Project Kickoff'* milestone.
- **Phase 2: *Design Phase*** (4 weeks): Following planning and analysis, the design phase begins and lasts for 4 weeks and concludes with the *'Design Approval'* milestone.

- **Phase 3:** ***Implementation Phase*** (16 weeks): The longest phase is implementation, spanning 16 weeks. It ends with the *'Feature Completion'* milestone, marking the completion of all planned features.
- **Phase 4:** ***Testing Phase*** (5 weeks): This phase involves three specific types of testing: unit testing, system testing, and user acceptance testing. The testing phase lasts for 5 weeks.
- **Phase 5:** ***Deployment Phase*** (3 weeks): The final phase is deployment, which lasts for 3 weeks and concludes with the *'Project Completion'* milestone.

10. Updated Budget

To develop the proposed system, the programmer/developer aims to minimize costs while still achieving the desired functionality. Thus, the budget allocation for the proposed system primarily involves the use of open-source software and software available on a free-trial basis, such as Python[5], Flask[6], Dash[7], AWS EC2[2], Amazon RDS[2], Amazon S3[2], and Amazon CloudWatch[2] during its development phase. Additionally, the existing computers and workstations within the school campus will serve as the hardware requirements, which is free-of-charge. As the system will integrate AI technology, specifically ChatGPT, the developer will utilize the free version, ChatGPT3.5, throughout the development and testing phase. However, it is recommended that the client (the school) upgrades to ChatGPT Plus during the deployment phase to provide the proposed system access to a larger number of tokens and advanced data analysis capabilities. The cost of this subscription is USD $20 per month[8].

11. Appendices

11.1 Definitions, Acronyms, Abbreviations

- **Attainment** – the level of achievement that the students have reached at a particular point in time, often measured through internal assessments.
- **Progress** – the improvement or development students make over a certain period. It's often measured by comparing students' attainment at different points in time.
- **External Benchmark Tests** – standardized assessments conducted by an outside agency. They provide a way to measure a student's performance against a wider population of students and identify where a student stands in comparison to a broader group.[4]
- **AWS** – Amazon Web Services, a service provided by the Amazon that uses distributed IT infrastructure to provide different IT resources available on demand
- **EC2** – Elastic Compute Cloud, a web service that provides secure, resizable compute capacity in the cloud.
- **t2.micro instance** - burstable performance instances that provide a baseline level of CPU performance with the ability to burst above the baseline.
- **Amazon S3** – Amazon Simple Storage Service, a scalable, high-speed, web-based cloud storage service designed for online backup and archiving of data and applications on AWS.
- **Amazon CloudWatch** - a monitoring service for Amazon Web Services cloud resources and applications run on AWS.
- **AWS Management Console** - a web application that comprises and refers to a broad collection of service consoles for managing AWS resources.
- **RHEL** – Red Hot Enterprise Linux, a linux server platform that allows customers to build and test enterprise applications on AWS and in their on-premises data centers.

- **SLES** – Suse Linux Enterprise Server, an adaptable and easy-to-manage linux server platform that allows developers and administrators to deploy business-critical workloads on Amazon EC2.

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